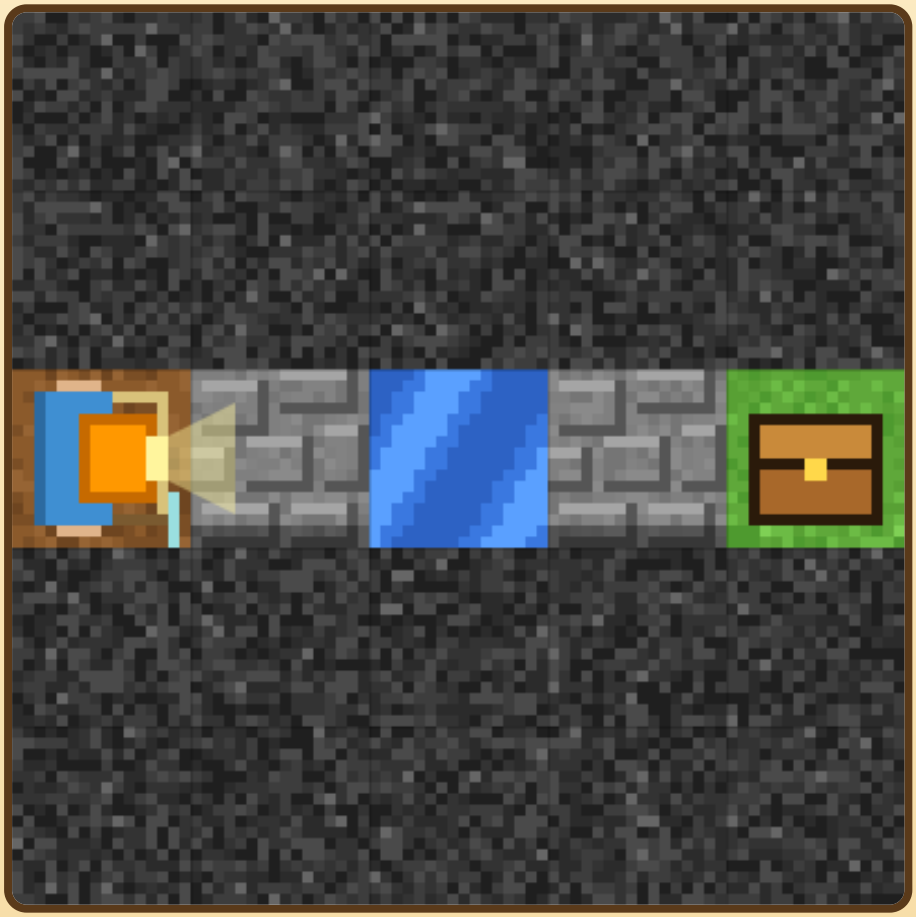
For kids and teachers

Learn Programming by Breaking & Building

From blocks to Python — 14 adventure stages



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How to use this book

Use this book together with the “Break & Build Quest” app on a computer or tablet. You will learn how to think like a programmer by building programs with blocks and with Python.

1. Read the **problem** and look at the picture.
2. Think about the order: trace it with your finger or write it in your notebook.
3. Build the program in the app and press “▶ Run”. If it doesn’t work, find what was different and fix it.
4. Stuck? Look at the 💡 **Hint**. When you are done, turn the page and check the 📖 **Answer** and the ✨ **Key point**.

There is more than one right answer. If it works, your program is correct, even if it is different from the book. Mistakes are hints — try again and again!

Apps in this book

App	Tasks	Grade
Break & Build Quest	7	Grade 4+
Python Break & Build	7	Grade 5 – junior high

PART 1

Part 1 The adventure book for kids

Chapter 1

Break, carry, place and go

Move the explorer to the treasure chest. Break the stone blocks in the way, and place blocks on water to build bridges.

Command	What it does
move forward / turn right, left	Move one square / change direction on the spot
⛏️ break the block in front	Breaks the stone in front. **You carry one more block.**
🧱 place a block in front	Places a block in front (on water it becomes a bridge). **You carry one less block.**
if the front is ○○	Checks whether the front is stone, water or an open path
repeat until goal	Repeats until you reach the chest

The big idea in this app

Remember **how many you carry**. With no blocks, you cannot build a bridge, so “break first, then place” matters. Computers keep numbers like this in **variables**.

When it doesn’t work

- Look at “Blocks you have” on the screen.
- On the worksheet, write how many you carry after each move.

Chapter 2

🧱 Break & Build Quest

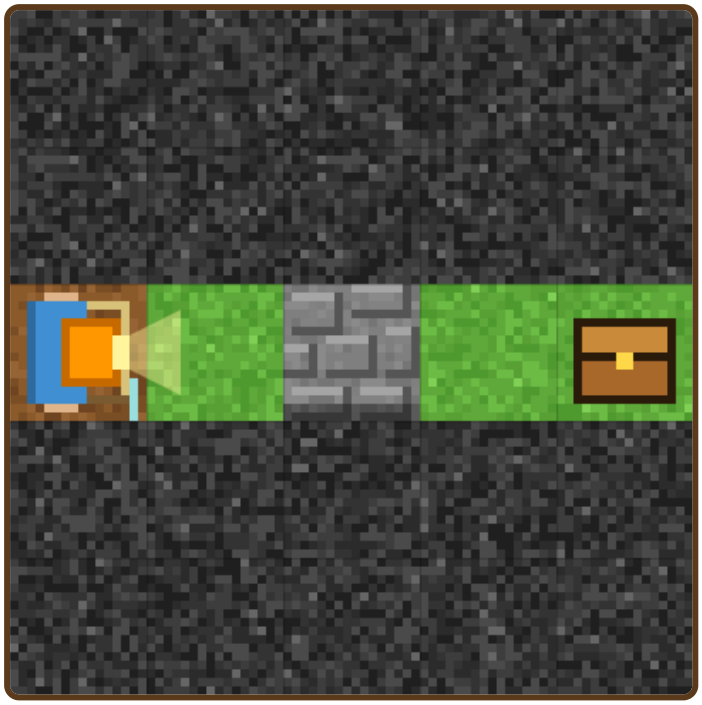
Break stone, carry it, and place it on water to build bridges. You will go from **order → loops → counting what you carry → if → two ifs**.

Commands in this chapter

move forward / turn right, left / ⛏️ break / 🧱 place / repeat ○ times / if the front is ○○ (else) / repeat until goal

Stage 1: Break and go

Sequence



The start

Problem

A stone block is in the way. Break it with "⛏️ break the block in front", then go to the treasure chest!

🎯 **Goal**：Use “break”

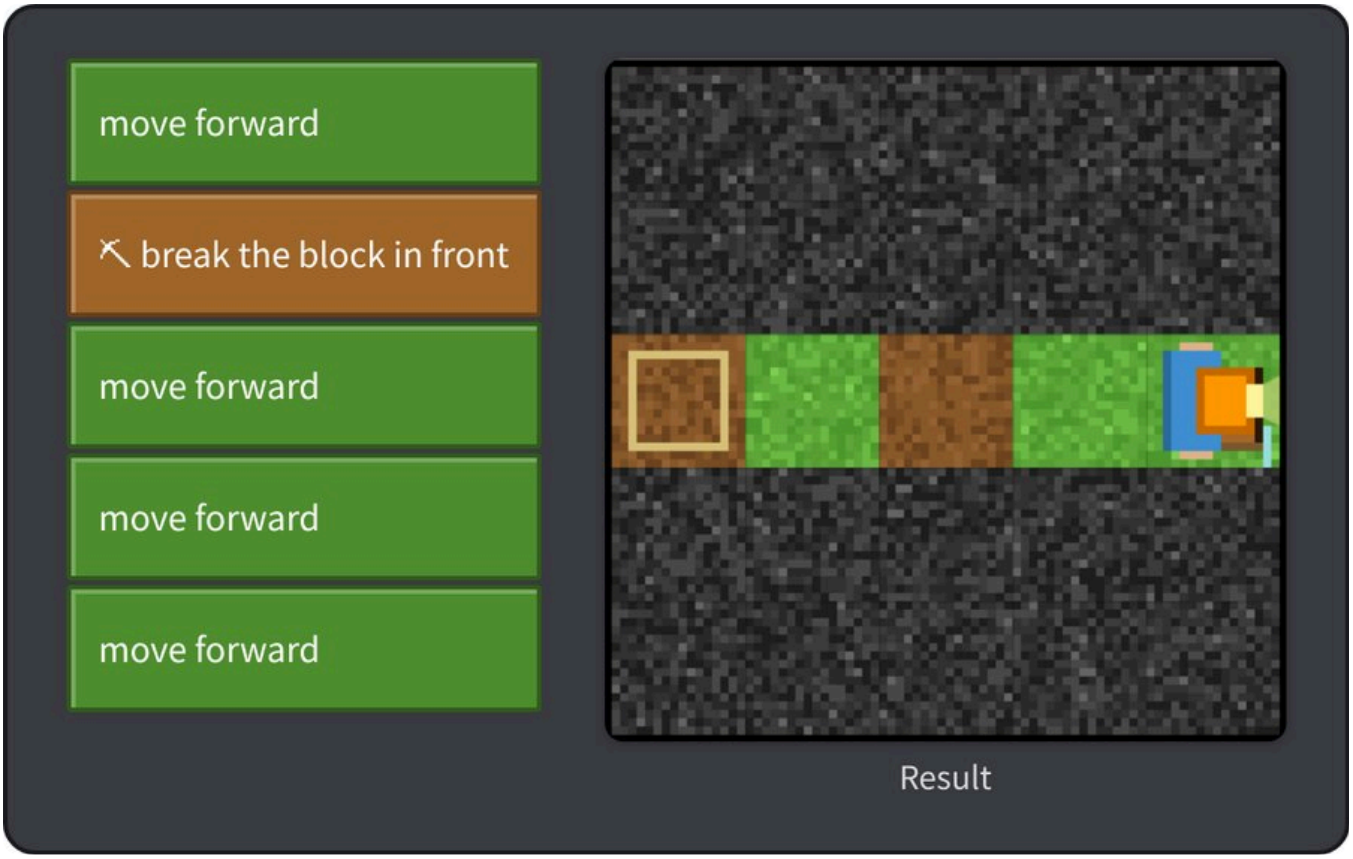
💡 **Hint (look if you get stuck)**

Forward 1 → break → forward 3

✏ **Write your own idea**

The answer is on the next page →

📖 Answer　Stage 1: Break and go



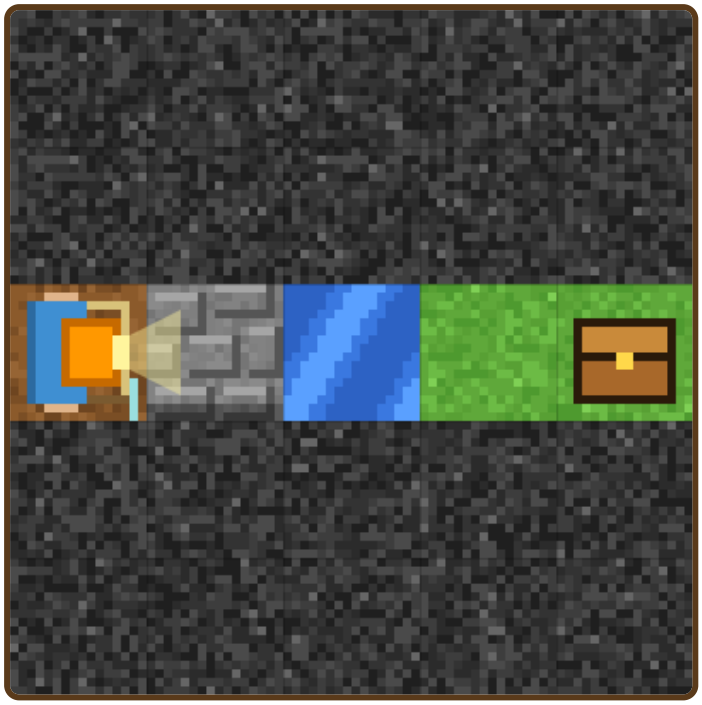
> ✨ **Key point**
>
> Break the stone in front, and it becomes a path.

> ⚠ **Common mistake**
>
> Moving without breaking bumps into the stone.

🎬 Video: `videos/en/craft-01.mp4`

Stage 2: Build a bridge

Sequence Counting



The start

Problem

There is water ahead. You need a block to make a bridge. Break the stone first — then you will have a block!

🎯 **Goal**：Break → hold → place

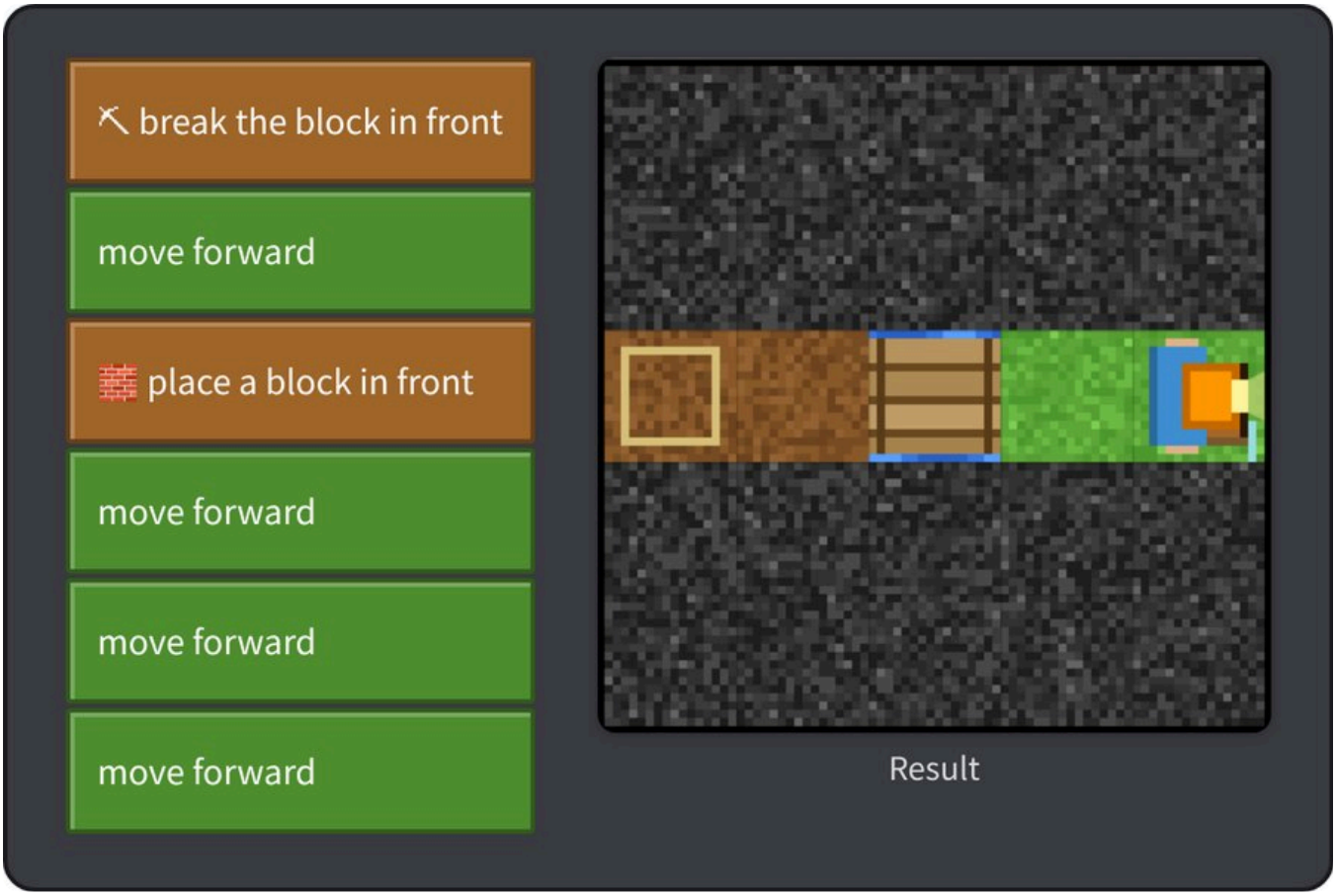
💡 **Hint (look if you get stuck)**

Break → forward → place → forward 3

✏ **Write your own idea**

The answer is on the next page →

📖 Answer　Stage 2: Build a bridge



✨ **Key point**

You start with no blocks. Break first, then place.

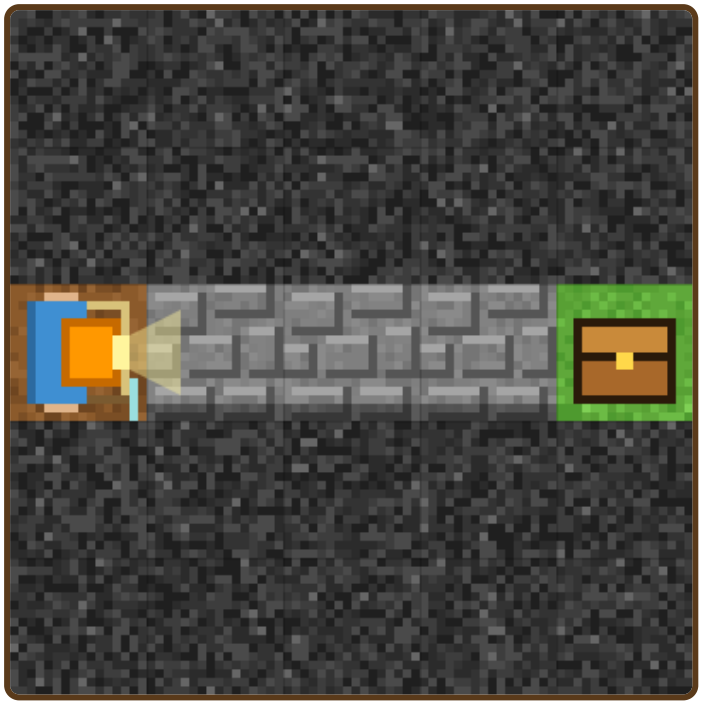
⚠ **Common mistake**

Placing first fails because you have no blocks yet.

🎬 Video: `videos/en/craft-02.mp4`

Stage 3: Break with a loop

Loops



The start

Problem

Stone blocks in a row! Put "break → move" into a loop.

🎯 Target: 4 blocks or fewer

🎯 **Goal**：Break with a loop

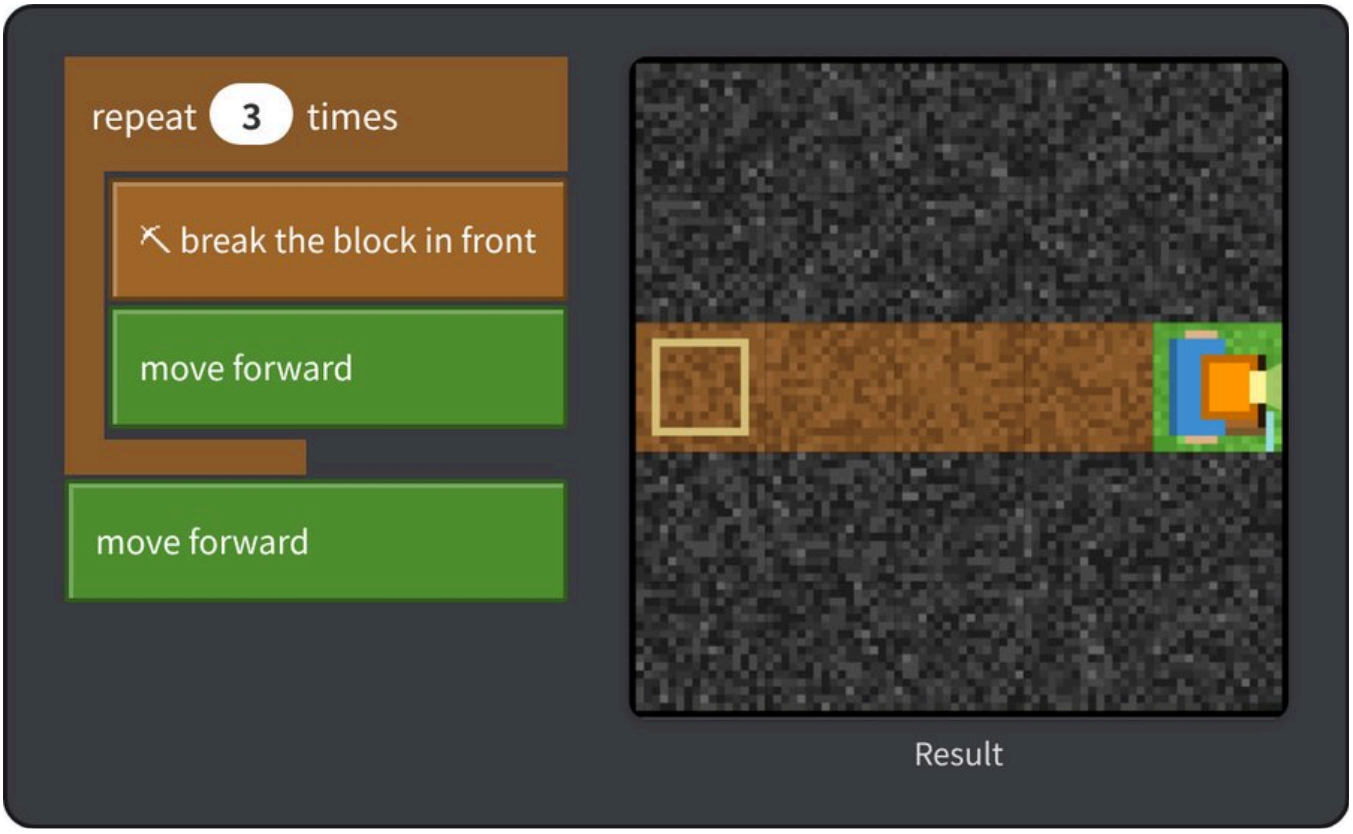
💡 **Hint (look if you get stuck)**

"repeat 3 (break → move forward)" → "move forward"

✏ **Write your own idea**

The answer is on the next page →

📖 Answer　Stage 3: Break with a loop



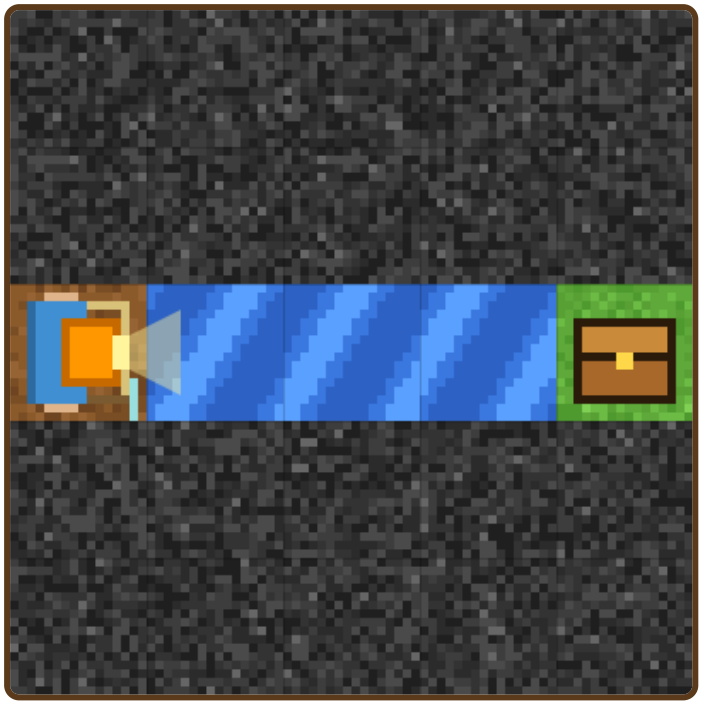
> ✨ **Key point**
>
> You repeated “break → move” 3 times.

> ⚠ **Common mistake**
>
> Moving first bumps into stone. Break first.

🎬 Video: `videos/en/craft-03.mp4`

Stage 4: A long bridge

Loops Counting



The start

Problem

You have 3 blocks. Make a long bridge over the river.

🎯 Target: 4 blocks or fewer

🎯 **Goal**：Use up the blocks you have

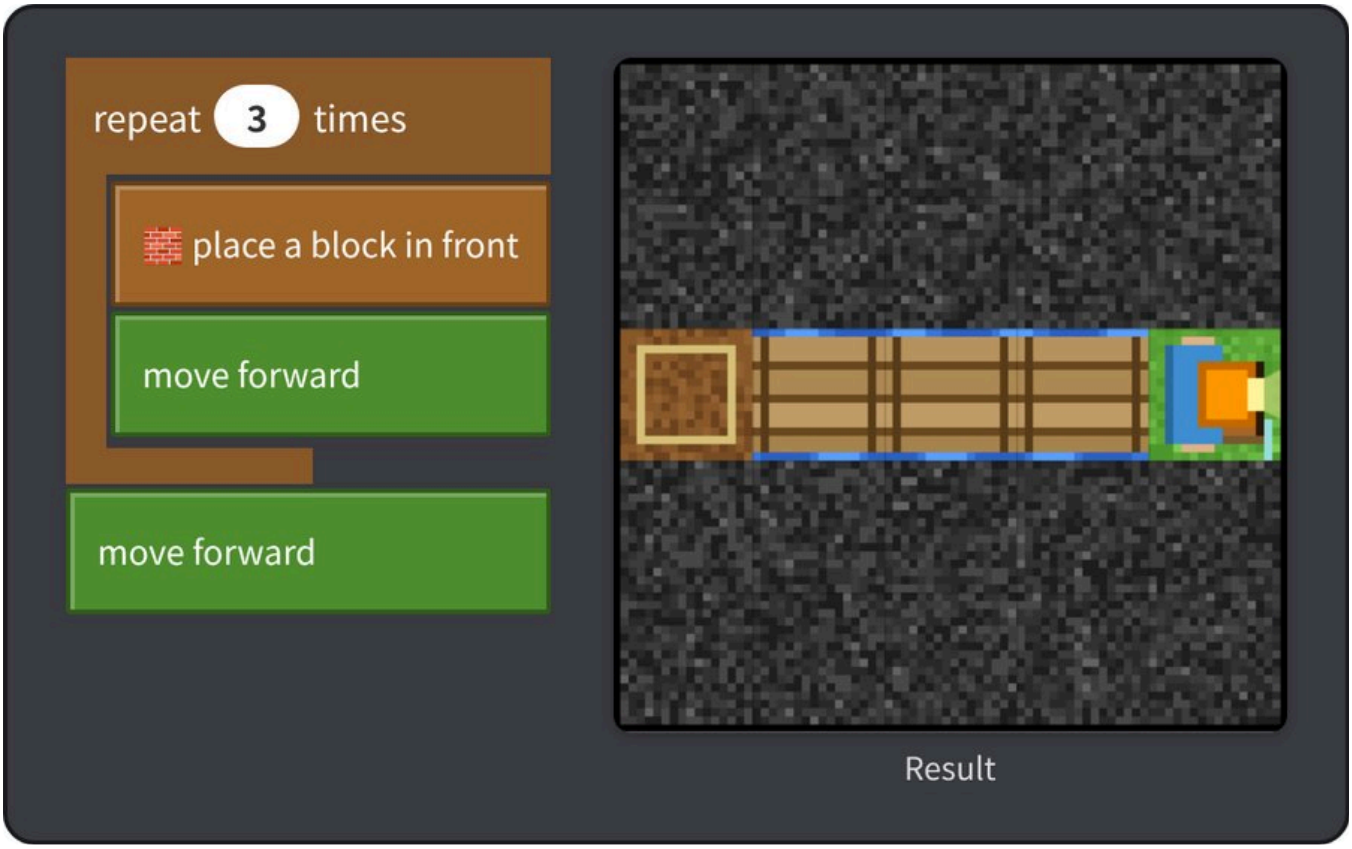
💡 **Hint (look if you get stuck)**

"repeat 3 (place → move forward)" → "move forward"

✏ **Write your own idea**

The answer is on the next page →

📖 Answer　Stage 4: A long bridge



✨ **Key point**

You had 3 blocks. After placing 3, you have exactly 0.

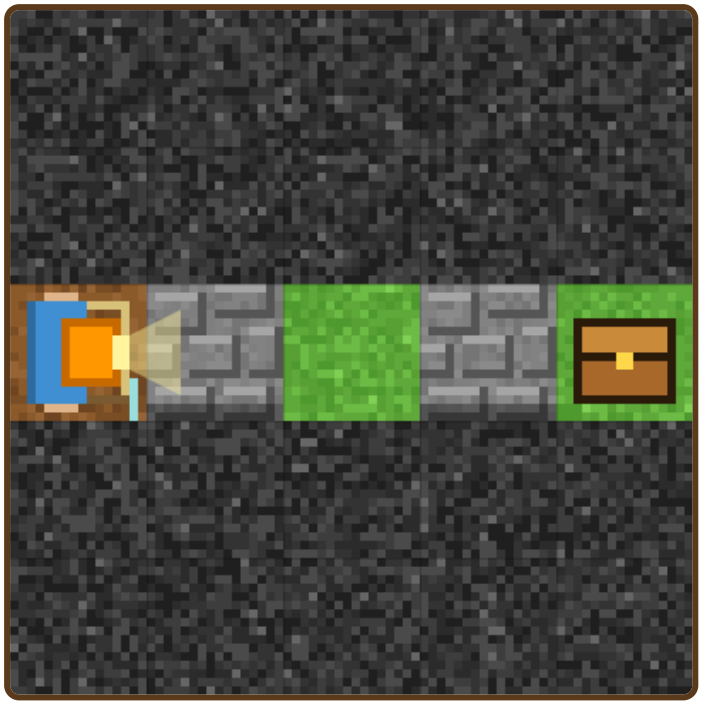
⚠ **Common mistake**

Moving before placing drops you into the water.

🎬 Video: `videos/en/craft-04.mp4`

Stage 5: If there is stone

Conditions Loops



The start

Problem

Some squares have stone and some don't. Breaking where there is no stone is a mistake! Use "if" to check first.

🎯 Target: 4 blocks or fewer

🎯 **Goal**：Check first, then break (if)

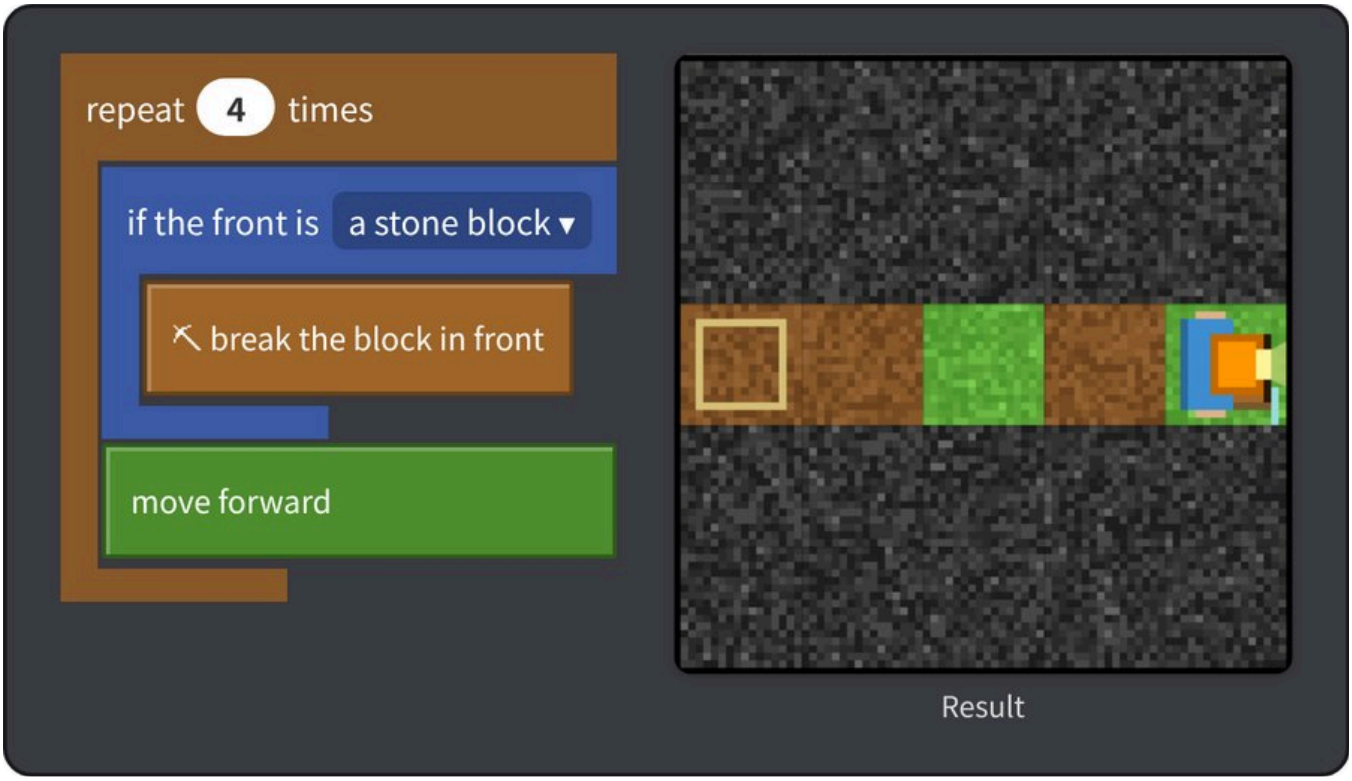
💡 **Hint (look if you get stuck)**

"repeat 4: if the front is a stone block, break it → move forward"

✏ **Write your own idea**

The answer is on the next page →

📖 Answer　Stage 5: If there is stone



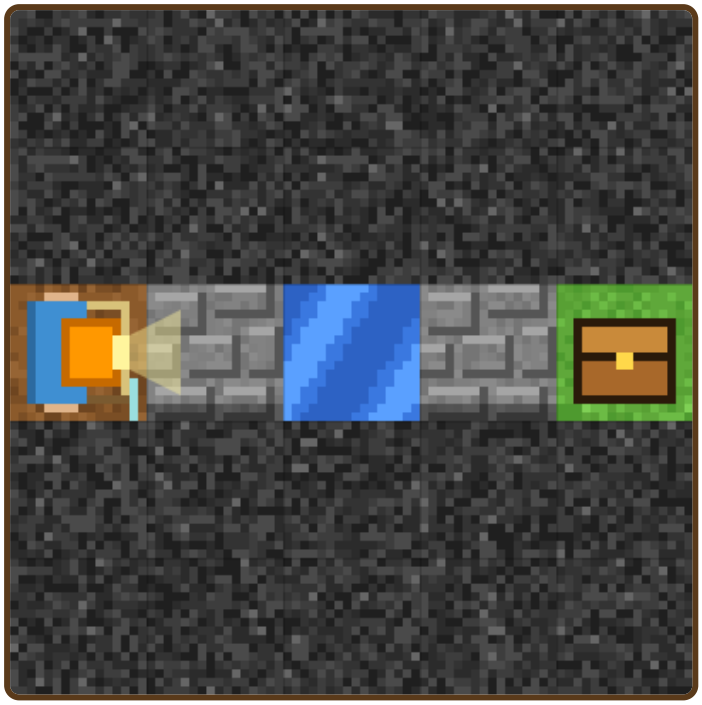
> ✨ **Key point**
>
> Break only when there is stone. Breaking where there is none is a mistake!

> ⚠ **Common mistake**
>
> Trying to break when there is no stone fails.

🎬 Video: `videos/en/craft-05.mp4`

Stage 6: Stone and water

Conditions Counting



The start

Problem

Stone and water are mixed. Break stone, bridge water. Breaking stone gives you a block, and placing one uses it up. Watch how many you have!

🎯 Target: 6 blocks or fewer

🎯 **Goal**：Stone and water: two “if”s

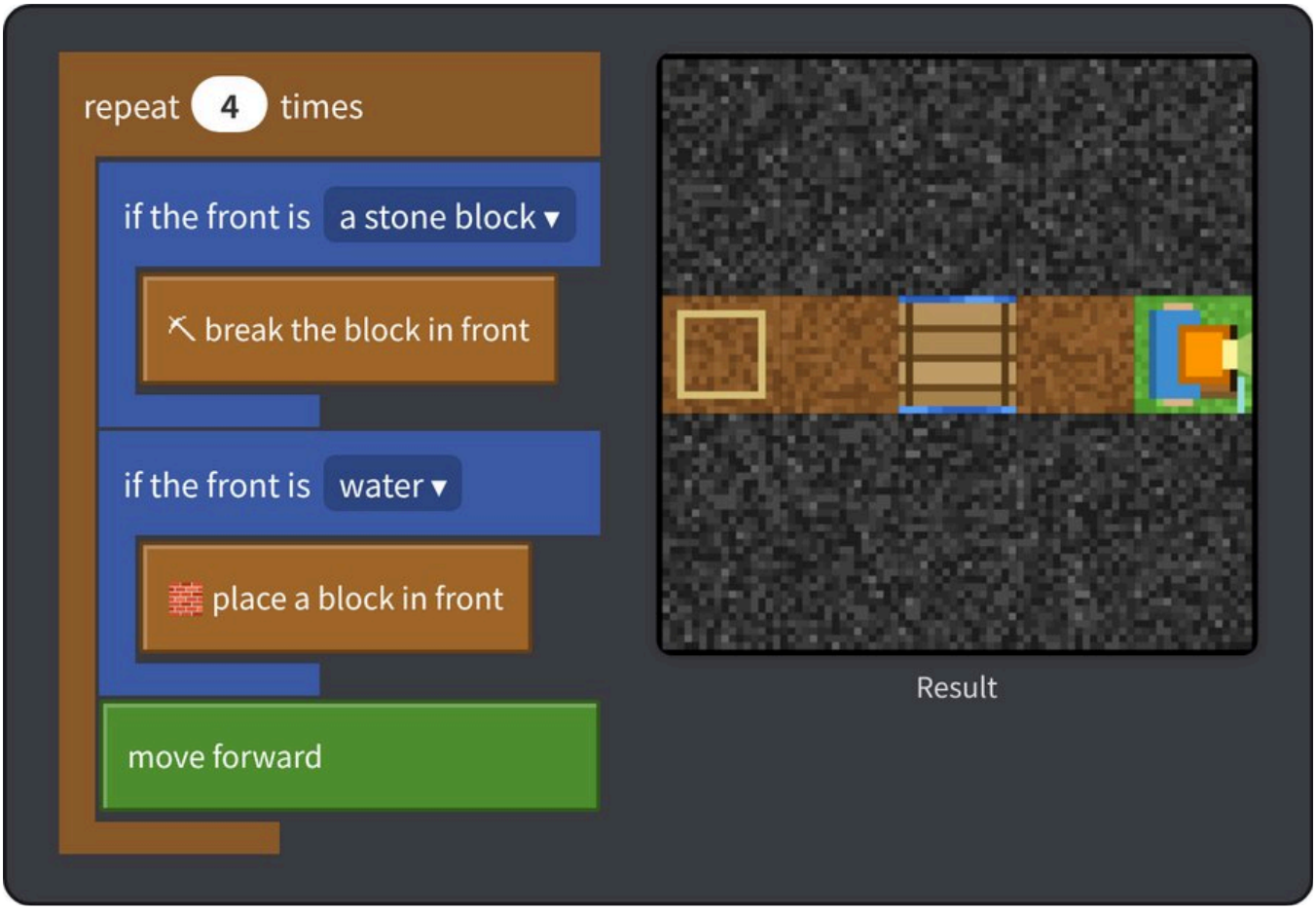
💡 **Hint (look if you get stuck)**

Repeat 4 times: if stone → break / if water → place / move forward

✏ **Write your own idea**

The answer is on the next page →

📖 Answer　Stage 6: Stone and water



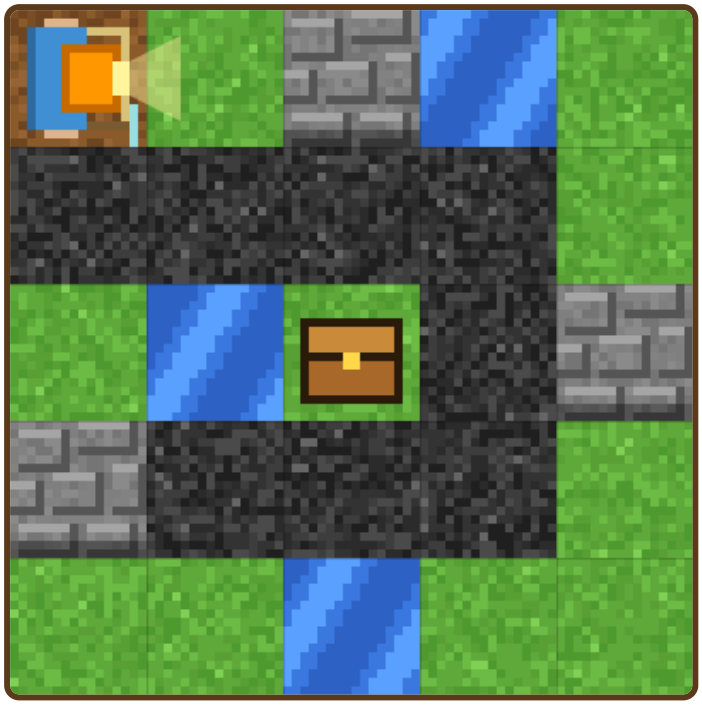
> ✨ **Key point**
>
> Breaking adds one, placing uses one: 0 → 1 → 0 → 1.

> ⚠ **Common mistake**
>
> Forgetting to check for water makes you fall in.

🎬 Video: `videos/en/craft-06.mp4`

Stage 7: Spiral cave

Conditions | Repeat until | Nesting



The start

Problem

A spiral cave full of stone and water. Use "repeat until goal" and decide what to do by checking the front.

🎯 Target: 8 blocks or fewer

🎯 **Goal**：Check the front as you go

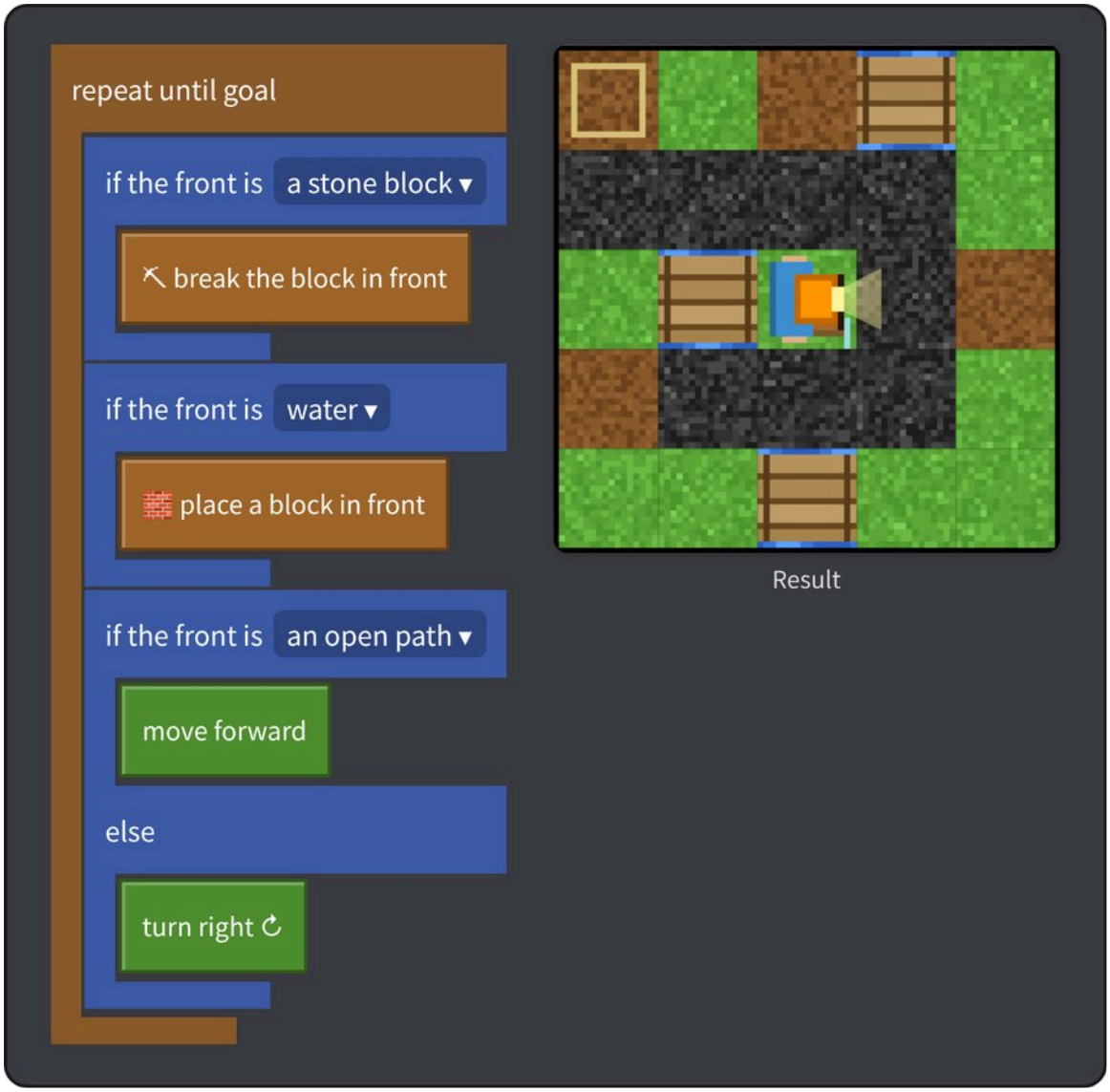
💡 **Hint (look if you get stuck)**

Inside the loop: if stone → break / if water → place / if the front is an open path → move, else turn right

✏ **Write your own idea**

The answer is on the next page →

📖 Answer　Stage 7: Spiral cave



> ✨ **Key point**
>
> Stone? Break. Water? Place. Open? Move. Otherwise, turn right!

> ⚠ **Common mistake**
>
> Hard dark rock cannot be broken. If you cannot move, turn right.

🎬 Video: `videos/en/craft-07.mp4`

Column

Count up and down

Breaking is +1, placing is −1. When stone and water are mixed, as in stage 6, counting “how many do I have now?” as you go shows where you would run out. Counting is an important skill for reading programs.

Chapter 3

🐍 Break & Build in Python

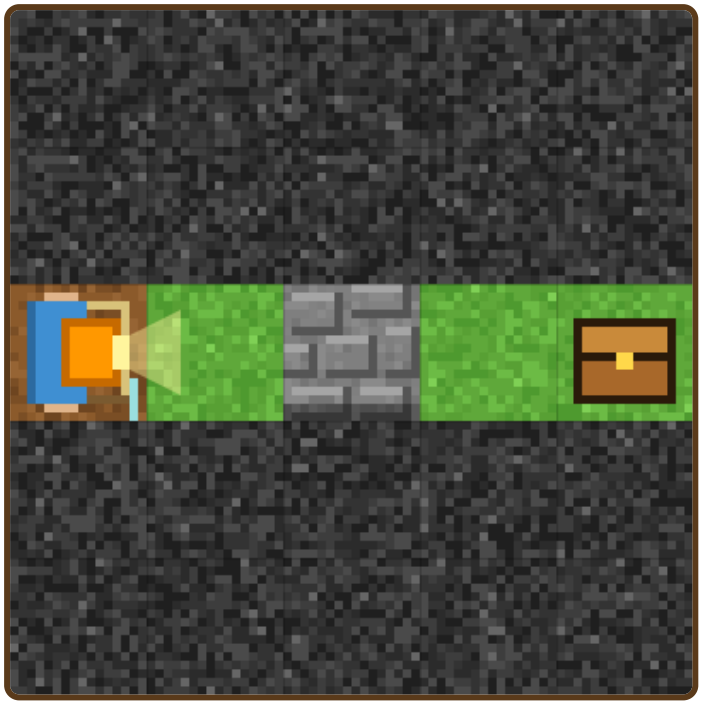
Write the adventure in **Python**: `mine()` breaks and `place()` places. Check with `if front_is('stone'):` and split into three or more cases with `elif`.

Commands in this chapter

forward() / turn_right() / mine() / place() / for i in range(3): / if front_is('stone'): / elif front_is('water'): / while not at_goal():

Stage 1: Break and go

Sequence | Python syntax (colon & indent)



The start

Problem

A stone block is in the way. mine() breaks the block in front of you. Break it and go to the treasure chest!

🎯 **Goal**：Use mine()

💡 **Hint (look if you get stuck)**

forward() → mine() → forward() ×3

✏ **Write your own idea**

The answer is on the next page →

📖 Answer　Stage 1: Break and go

```
forward()
mine()
forward()
forward()
forward()
```

✨ Key point

mine() breaks the stone in front, and it becomes a path.

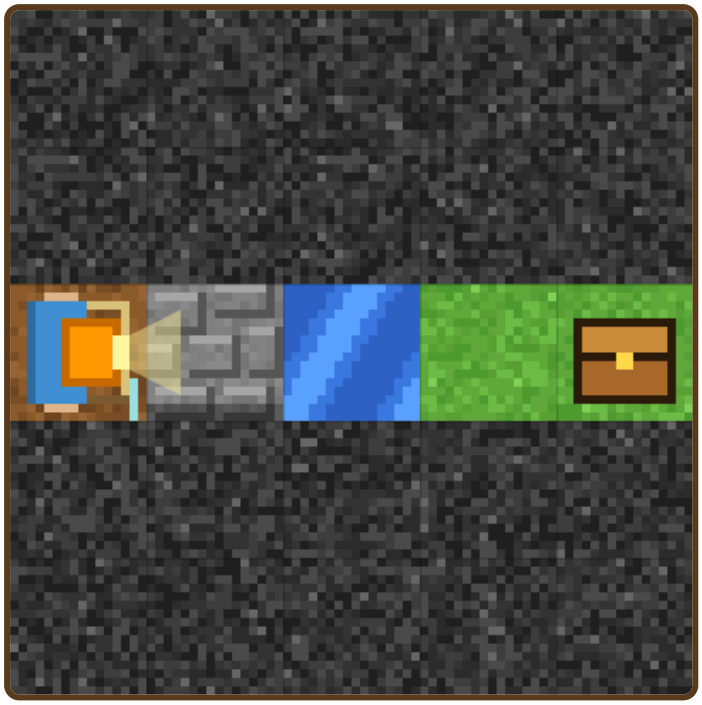
⚠ Common mistake

mine() where there is no stone fails. Move forward once before breaking.

🎬 Video: `videos/en/pycraft-01.mp4`

Stage 2: Build a bridge

Sequence | Python syntax (colon & indent)



The start

Problem

place() puts a block in front of you. On water it becomes a bridge. Break the stone first to get a block!

🎯 **Goal**：The order: mine() → place()

💡 **Hint (look if you get stuck)**

mine() → forward() → place() → forward() ×3

✏ **Write your own idea**

The answer is on the next page →

📖 Answer　Stage 2: Build a bridge

```
mine()
forward()
place()
forward()
forward()
forward()
```

✨ Key point

You start with no blocks. mine() to get one, then place() to build a bridge.

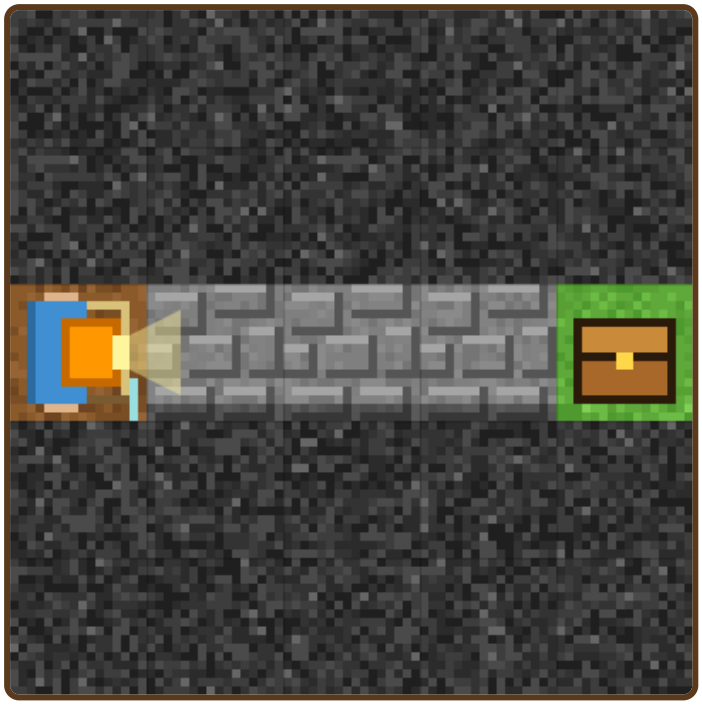
⚠ Common mistake

forward() first bumps into the stone. Move to just before the water before place().

🎬 Video: `videos/en/pycraft-02.mp4`

Stage 3: Break with for

Loops | Python syntax (colon & indent)



The start

Problem

Stone blocks in a row! Repeat “mine() → forward()” with for.

🎯 **Goal**：Break with for

💡 **Hint (look if you get stuck)**

for i in range(3): mine(), forward() → forward()

✏ **Write your own idea**

The answer is on the next page →

📖 Answer　Stage 3: Break with for

```
for i in range(3):
    mine()
    forward()
forward()
```

✨ **Key point**

“mine() → forward()” 3 times. The last forward() is outside the for (not indented).

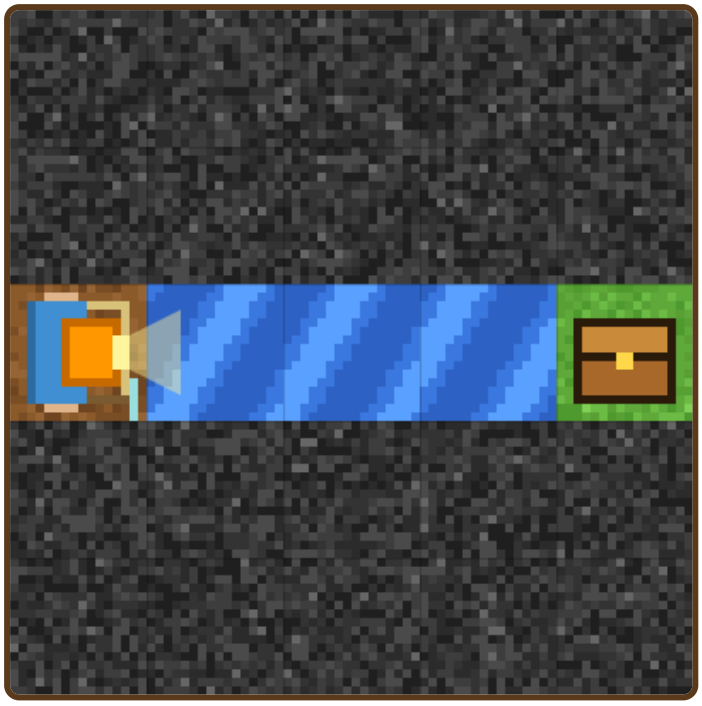
⚠ **Common mistake**

If forward() is not indented, mine() runs 3 times in a row and fails.

🎬 Video: `videos/en/pycraft-03.mp4`

Stage 4: A long bridge

Loops | Counting | Python syntax (colon & indent)



The start

Problem

You have 3 blocks. Repeat “place() → forward()” to make a long bridge.

🎯 **Goal**：Use up the blocks you have

💡 **Hint (look if you get stuck)**

for i in range(3): place(), forward() → forward()

✏ **Write your own idea**

The answer is on the next page →

📖 Answer Stage 4: A long bridge

```
for i in range(3):
    place()
    forward()
forward()
```

✨ **Key point**

You have 3 blocks, so after 3 place() calls you have exactly 0.

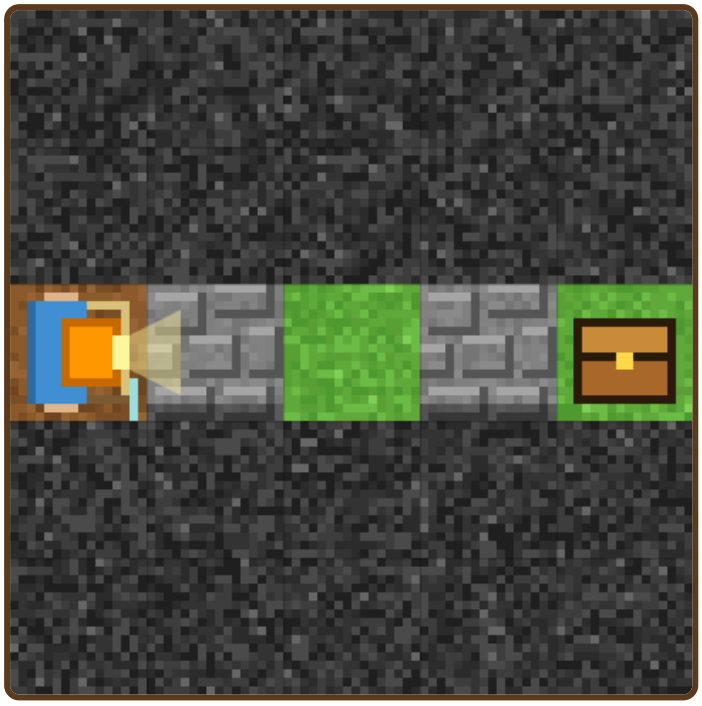
⚠ **Common mistake**

With range(4), the 4th place() runs out of blocks.

🎬 Video: `videos/en/pycraft-04.mp4`

Stage 5: if front_is('stone'):

Conditions | Loops | Nesting | Python syntax (colon & indent)



The start

Problem

front_is('stone') checks if the front is a stone block. Use if so you only break where there is stone.

🎯 **Goal**：Check with if front_is('stone'):

💡 **Hint (look if you get stuck)**

for i in range(4): if front_is('stone'): mine() → forward()

✏ **Write your own idea**

The answer is on the next page →

📖 Answer　Stage 5: if front_is('stone'):

```
for i in range(4):
    if front_is('stone'):
        mine()
    forward()
```

✨ Key point

front_is('stone') is True if the front is stone. Indent one more level inside if.

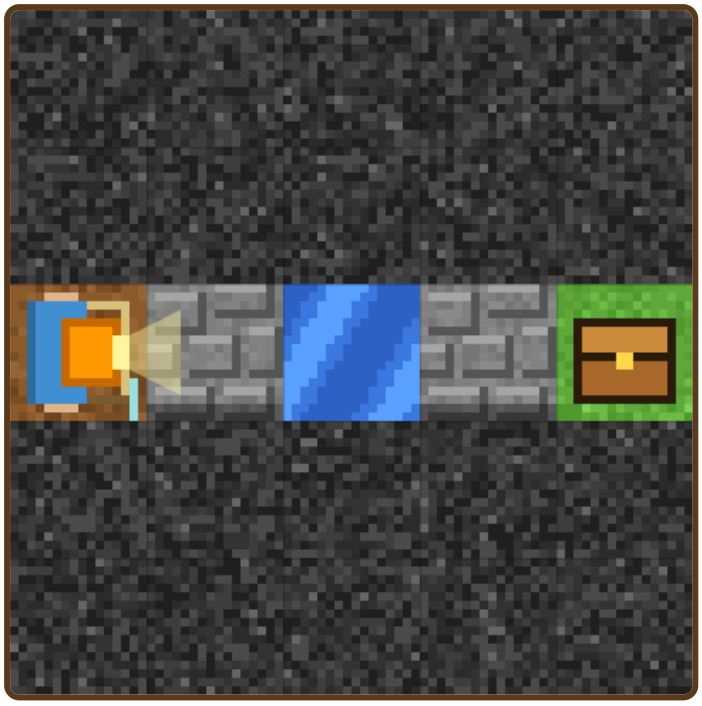
⚠ Common mistake

"stone" with double quotes is OK, but other words like 'rock' cause a ValueError.

🎬 Video: `videos/en/pycraft-05.mp4`

Stage 6: Stone and water with elif

Conditions | Loops | Nesting | Counting | Python syntax (colon & indent)



The start

Problem

Stone and water are mixed. If it is stone, mine(); elif it is water, place(). Breaking gives you a block, and placing uses one.

🎯 **Goal**：Split into two cases with elif

💡 **Hint (look if you get stuck)**

for i in range(4): if front_is('stone'): mine() / elif front_is('water'): place() / forward()

✏ **Write your own idea**

The answer is on the next page →

📖 Answer　Stage 6: Stone and water with elif

```
for i in range(4):
    if front_is('stone'):
        mine()
    elif front_is('water'):
        place()
    forward()
```

✨ Key point

Stone → mine(), else if water → place(). Breaking adds one and placing uses one (0 → 1 → 0 → 1).

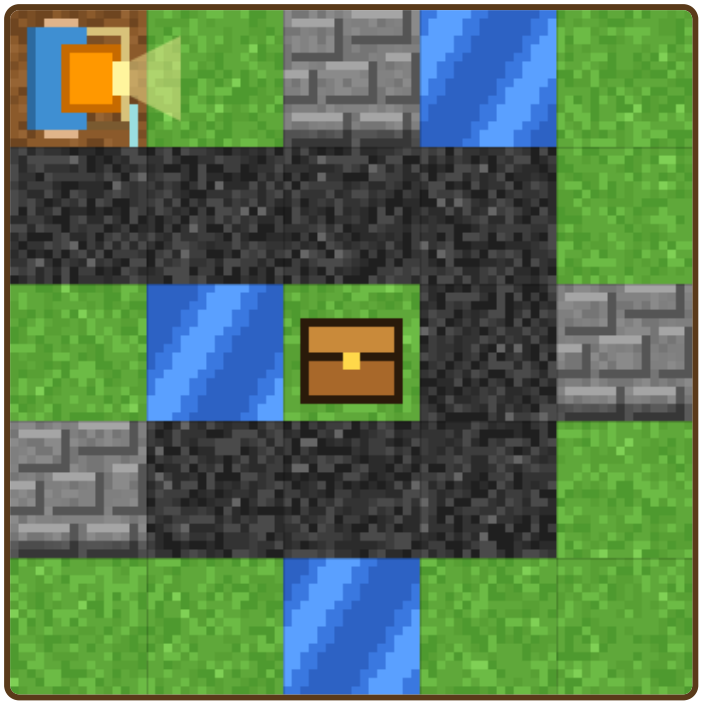
⚠ Common mistake

Writing “else if” causes a SyntaxError. Write elif at the same depth as if.

🎬 Video: `videos/en/pycraft-06.mp4`

Stage 7: Spiral cave

Conditions | Repeat until | Nesting | Generalization | Python syntax (colon & indent)



The start

Problem

Use “while not at_goal():” and decide by checking the front: stone → mine(), water → place(), path → forward(), otherwise turn_right().

🎯 **Goal**：Check the front as you go

💡 **Hint (look if you get stuck)**

while not at_goal(): if front_is('stone') … elif front_is('water') … elif front_is('path') … else: turn_right()

✏ **Write your own idea**

The answer is on the next page →

📖 Answer Stage 7: Spiral cave

```
while not at_goal():
    if front_is('stone'):
        mine()
    elif front_is('water'):
        place()
    elif front_is('path'):
        forward()
    else:
        turn_right()
```

✨ Key point

if, elif, elif and else split it into 4 cases. You check the front again on every loop.

⚠ Common mistake

Hard dark rock is not 'stone' for front_is. When you cannot move, else turns right.

🎬 Video: `videos/en/pycraft-07.mp4`

Column

How to use elif

`if` → `elif` → `else` checks from the top and runs only the first one that matches. Like “if stone, break; else if water, place; otherwise move”, you can split into three or more cases. Don’t forget the quotes in `'stone'`.

Well done!

You worked hard to the end! Let’s look back at the ideas you used.

Idea	Where did it appear?
Order	Break → carry → place (stages 1–2)
Loops	Pack “break → move” together (stages 3–4)
Counting what you carry	Up and down (stages 2, 4, 6)
If (conditions)	Break only stone, place on water (stages 5–6)
Repeat until	Check as you go to the goal (stage 7)
Python	Split cases with if / elif / else

When you clear every stage, get your certificate from “⚙ Settings” → “📜 Certificate”.

PART 2

Part 2 Guide for teachers and parents

About this guide

This book covers “Break & Build Quest” (7 stages) and “Python Break & Build” (7 stages). Part 1 is written in simple language for students.

Breaking stone adds one block; placing on water takes one away. Because the count (state) affects later commands, order really matters, which builds a basis for variables. Later stages combine cases (stone, water, path) with conditions.

Aims

- Plan the order of steps while thinking about how the count changes.
- Build programs that choose actions under several conditions.
- In Python, learn multi-way branching with if / elif / else.

Overview of the stages

App	#	Stage	Ideas
🧱 Break & Build Quest	1	Break and go	Sequence
	2	Build a bridge	Sequence, Counting
	3	Break with a loop	Loops
	4	A long bridge	Loops, Counting
	5	If there is stone	Conditions, Loops
	6	Stone and water	Conditions, Counting
	7	Spiral cave	Conditions, Repeat until, Nesting
🐍 Python Break & Build	1	Break and go	Sequence, Python syntax (colon & indent)
	2	Build a bridge	Sequence, Python syntax (colon & indent)
	3	Break with for	Loops, Python syntax (colon & indent)
	4	A long bridge	Loops, Counting, Python syntax (colon & indent)
	5	if front_is('stone'):	Conditions, Loops, Nesting, Python syntax (colon & indent)
	6	Stone and water with elif	Conditions, Loops, Nesting, Counting, Python syntax (colon & indent)
	7	Spiral cave	Conditions, Repeat until, Nesting, Generalization, Python syntax (colon & indent)

Suggested order and teaching points

Students	Stages	Lessons	Key ideas
Grades 4–5	1–4	1–2	Order, loops, counting up and down
Grades 5–6	5–7	2	Conditions (two cases), repeat until
Grade 6 – junior high	Python 1–7	2–3	if / elif / else, while

Common difficulties and support

- **Placing with nothing to place:** in stage 2, use the on-screen count to show you must break first.
- **Keeping count:** have students write the count after each move in the Worksheet 5 table.
- **Two conditions (stage 6):** have students say “when, and what to do” in words before building.
- **Python:** watch the quotes in `front_is('stone')` and the order of `elif` (checked from the top).

Example 45-minute lesson (stages 4–6)

Time	Activity	Support
Intro 5 min	Confirm that breaking adds and placing takes away; share the goal.	Point to the on-screen count.
Main 30 min	Build a long bridge; break only stone; split actions for stone and water.	Record the count in a table.
Wrap-up 10 min	Summarize the value of counting and cases.	Introduce the word “variable”.

Assessment (example)

Area	Meets expectations (B)	Exceeds (A), e.g.
Knowledge & skills	Combines blocks (code) correctly and completes the task.	Uses loops and conditions to complete it with a short, clear program.
Thinking, judgment & expression	Finds where the error is from the result and fixes it.	Explains the rule or steps in words and makes a program that works in other cases too.
Attitude toward learning	Keeps trying even when it doesn’t work.	Shares ideas with friends and looks for better ways on their own.

Use the worksheets (1 reflection; 2, 5, 7 etc.) and the app progress (8 check card, certificate) as evidence.

Using the apps and related materials

Requirements and tips

- Works in the browser on computers, tablets and smartphones. No installation needed. An internet connection is required.
- Cleared stages are saved in that device’s browser. If devices are shared, check progress at the start of each lesson.
- **Teacher mode:** add `?sensei` to the end of the address to show an “Answer” button.
- From “⚙ Settings” you can switch Japanese/English, use the pixel font, switch to the Python version and show the certificate.

Related materials

Material	Contents
Guide (guide.html)	Goals, common mistakes, hints and answers for every task. This book is built from the same data.
Worksheets (PDF)	5 Space / Break & Build (order and count table), 7 Python, 1 goal & reflection, 8 check card.
Lesson plans (Japanese)	Example 45-minute lessons (open them from each task in the Guide).
Videos (videos.html)	Short videos showing how to solve each task.
Lesson slides	04 Break & Build Quest, 08 Python Break & Build (PowerPoint).